

Technical Report Writing (TRW)

Chapter No 5
(Techniques)

Researching

Chapter In a Nutshell

- Purpose of research
- Questioning; The Basic Skill of Researching
- Collecting Information from People
- Collecting Published Information

Research and its Purpose

Research is the collection of relevant data/information to be used for the production of the document. Research forms the basis of every document. A technical writer cannot write a document without proper research. We have problems because we have wants and needs. In order to solve the problems, the collection of relevant data is essential.

Questioning; How to formulate questions?

The fundamental concept of research is questioning. The answers which we get for our intelligently designed questions provide us the relevant facts. Questioning is of two types:

- 1) Close-ended Questions/Close Questions
- 2) Open-ended Questions/Open Questions

Close Ended Questions

The nature of close questions is that they generate a limited/specific and to the point response from the respondent. Therefore, closed questions are used to obtain very basic sort of information and nature of these questions is objective.

Open Ended Questions

An open question has such a nature that it allows a detailed and more involved response from the respondent. Therefore, these are used to obtain detailed information/secondary information. Hence, their nature is of subjectivity or they are contrary to the objective nature of closed questions.

Collecting Information from People

A researcher may utilize the following sources for collecting relevant information

- Interview
- Survey
- Observe
- Test
- Read

Interviewing

An interview is conducted with one respondent at a time. Therefore, it is time-consuming. However, it provides the researcher with detailed information and very close observation. In order to conduct interviews, a researcher must interact flexibly and design questions intelligently. The goal is to discover the appropriate facts from a person who knows them. To conduct an effective interview, you must:

- Prepare carefully

- Maintain a professional attitude

- Probe

- Record

Probing

Most people know more than they say in their initial answers, so you must be able to get at the material that has been left unsaid. A researcher may encounter a situation in which the respondents do not provide adequate information. Normally, it has been felt that the respondents say less as compared to what they know in their first response. In such situations, the following techniques may be utilized by the researcher in order to get the unsaid information in the first response:

Ask open ended questions

Use the echo technique

Reformulate

Ask for a process description

Echo Technique

(repeating significant words)

The researcher notes the first response of the respondent and asks him to explain one important jargon/terminology or a phrase. The response provides probing to the researcher because now the response will be more involved and detailed.

Reformulation

(repeating in your own words)

In this case, the researcher either rephrases the first response of the respondent or gives a comment on the first response of the respondent. As a result, the respondent indulges in giving clarifications/explanations and the researcher gets the relevant information in a more detailed way.

Surveying

- On contrary to interviews, surveys are less time taking because in surveys, the researcher can engage a lot of respondents at one time. Surveys consist of written responses that are obtained on a written form/questionnaire (survey-form). This questionnaire must be designed very intelligently in order to get the exact/relevant information needed by the researcher.
- A survey form may consist of both close-ended and open-ended questions, but the open-ended questions and their responses are difficult to tabulate and secondly, surveys are conducted to obtain basic sort of information, hence close ended questions are used in a survey.
- Use context-setting introductions. The questionnaires/survey-form must contain a context-setting introduction at the top. It provides the respondents answers to:
 - a) Why the researcher needs this information.
 - b) How will this information be utilized by the researcher?
 - c) Why the researcher has chosen the respondent for this survey.
- The questions must be designed carefully.
- The questionnaire/form should be made in such a way that it provides facility to the researcher, when he collects/tabulates information from the form. It's language should be easy to understand and respond to for the respondents.

Observing & Testing

Observing

Observing is watching intently. You place yourself in the situation to observe and record your observations. When you observe to collect information, you do so with the same questions in mind as when you interview: What are the basic facts? What is their significance?

Testing

To test is to compare items in terms of some criterion or set of criteria

Typical questions might be:

- Which one produces 100 copies faster?
- Which one makes clearer back to back copies?
- Which one generates the least heat?

After determining suitable questions, you have people use both machines and then record their answers to your questions.

Collecting Published Information

- Develop a search strategy
- Determine Your audience
- Generate Questions
- Predict Probable sources
- Consider the age of the Information
- Consider the technical level of the Information
- Search for “Keys”
- Find Keywords
- Watch for Key Documents